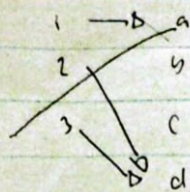


(26)

Studen, Ju SBIT-3E DISCRETE MATH 11/12/25

Determine the ff. relations if they are a function, injective, surjective and bijective.

1.) Domain = $\{1, 2, 3\}$ Codomain = $\{a, b, c, d\}$ where $A = \{(1, a), (2, d), (3, d)\}$

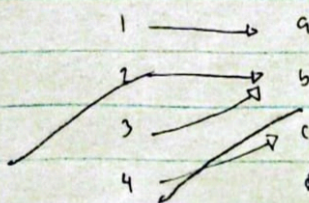


function = $\checkmark$
~~surjective~~ = $\times$
~~injective~~ = $\times$
~~bijective~~ = $\times$

set = $\checkmark$
 relation = $\checkmark$

(5)

2.) $B = \{(1, a), (2, b), (3, b), (4, c)\}$

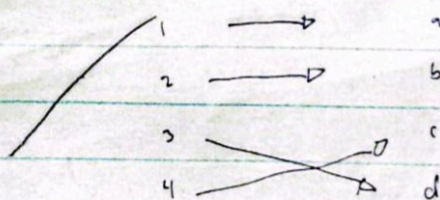


function = $\checkmark$
~~surjective~~ = $\checkmark$
~~injective~~ = $\times$
~~bijective~~ = $\times$

set = $\checkmark$
 relation = $\checkmark$

(5)

3.) $C = \{(1, a), (2, b), (3, d), (4, c)\}$

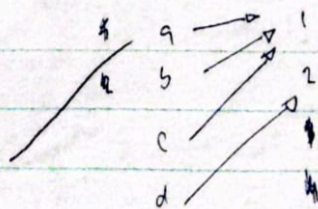


function = $\checkmark$
~~surjective~~ = $\checkmark$
~~injective~~ = $\checkmark$
~~bijective~~ = $\checkmark$

set = $\checkmark$
 relation = $\checkmark$

(5)

4.) $D = \{(a, 1), (b, 1), (c, 1), (d, 2)\}$



function = $\checkmark$
~~surjective~~ = $\checkmark$
~~injective~~ = $\times$
~~bijective~~ = $\times$

set = $\checkmark$
 relation = $\checkmark$

(5)

4.1. domain = $\{a, b, c, d\}$

4.2. codomain = $\{1, 2\}$

4.3. range = $\{1, 2\}$

4.4. preimage of 2 = d

4.5. $f^{-1}(\{1\}) = \{a, b, c\}$

4.6. $f(d) = 2$

(6)